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The effects of feeding α-amylase-enhanced corn silage with different dietary starch concentrations to lactating dairy cows on milk production, nutrient digestibility, and blood metabolites

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ABSTRACT

Corn silage is one of the most common ingredients fed to dairy cattle. Advancement of corn silage genetics has improved nutrient digestibility and dairy cow lactation performance in the past. A corn silage hybrid with enhanced endogenous α-amylase activity (Enogen, Syngenta Seeds LLC) may improve milk production efficiency and nutrient digestibility when fed to lactating dairy cows. Furthermore, evaluating how Enogen silage interacts with different dietary starch content is important because the ruminal environment is influenced by the amount of rumen fermentable organic matter consumed. To evaluate the effects of Enogen corn silage and dietary starch content, we conducted an 8-wk randomized complete block experiment (2-wk covariate period, 6-wk experimental period) with a 2 × 2 factorial treatment arrangement using 44 cows (n = 11/treatment; 28 multiparous, 16 primiparous; 151 ± 42 d in milk; 668 ± 63.6 kg of body weight). Treatment factors were Enogen corn silage (ENO) or control (CON) corn silage included at 40% of diet dry matter and 25% (LO) or 30% (HI) dietary starch. Corn silage used in CON treatment was a similar hybrid as in ENO but without enhanced α-amylase activity. The experimental period began 41 d after silage harvest. Feed intake and milk yield data were collected daily, plasma metabolites and fecal pH were measured weekly, and digestibility was measured during the first and final weeks of the experimental period. Data were analyzed using a linear mixed model approach with repeated measures for all variables except for body condition score change and body weight change. Corn silage, starch, week, and their interactions were included as fixed effects; baseline covariates and their interactions with corn silage and starch were also tested. Block and cow served as the random effects. Plasma glucose, insulin, haptoglobin, and serum amyloid A concentrations were unaffected by treatment. Fecal pH was greater for cows fed ENO versus CON. Dry matter, crude protein, neutral detergent fiber, and starch digestibility were all greater for ENO than CON during wk 1, but differences were less by wk 6. The HI treatments depressed neutral detergent fiber digestibility compared with LO. Dry matter intake (DMI) was not affected by corn silage but was affected by the interaction of starch and week; in wk 1, DMI was similar but by wk 6, cows fed HI had 1.8 ± 0.93 kg/d less DMI than LO cows. Milk, energy-corrected milk, and milk protein yields were 1.7 ± 0.94 kg/d, 1.3 ± 0.70 kg/d, and 65 ± 27 g/d greater for HI than LO, respectively. In conclusion, ENO increased digestibility but it did not affect milk yield, component yields, or DMI. Increasing dietary starch content improved milk production and feed efficiency without affecting markers of inflammation or metabolism.
Key words: forage, acute phase protein, carbohydrate, feed efficiency

INTRODUCTION

Corn silage is arguably the most important ingredient in North American dairy cattle diets. In the late 1990s nutritionists responded that corn silage was fed to 40–80% of lactating dairy cattle depending on region of the United States (Mowrey and Spain, 1999), and a similar survey found that 91% of lactating dairy cow diets and 82% of dry cow diets in the United States contain corn silage (Kellogg et al., 2001). Area harvested for corn silage in the United States has increased from 2.5 million hectares in 2018 to 2.7 million hectares in 2020 (USDA, 2021). Clearly, improving corn silage nutritive value could have substantial effects on dairy production in the United States.

An engineered corn trait providing enhanced endogenous α-amylase activity in the endosperm (Cueva et al., 2021b), which was originally designed for ethanol production, has increased feed efficiency when fed to beef cattle as dry rolled corn (Volk et al., 2021). When fed to dairy cattle as corn silage, there is evidence that it may increase milk yield by 2 to 3 kg/d, though it had

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less of an effect when evaluated on an ECM basis (Rebelo et al., 2020; Cueva et al., 2021b). Further research on this trait in different dietary formulations will be needed to elucidate whether it consistently improves the milk yield and feed efficiency of lactating dairy cows.

What has yet to be investigated is whether α-amylase-enhanced corn silage (Enogen, Syngenta Seeds LLC) interacts with the concentration of starch fed in the total mixed ration. One previous experiment investigating exogenous amylase fed at different dietary starch contents found that in low starch diets (~25% dietary starch) the exogenous amylase enzyme increased NDF digestibility (Weiss et al., 2011) but did not affect milk yield. It may be that in diets with >25% dietary starch the addition of amylase is unwarranted, but this has not been investigated with α-amylase-enhanced corn silage. Furthermore, increasing the dietary starch content from 21 to 31% of the diet by replacing forages with ground or high-moisture corn increased DMI, reduced total chewing time, reduced ruminal pH, and increased milk yield but reduced milk fat yield in peak lactation cows (Oba and Allen, 2003a). Feeding increased amounts of starch also tends to promotes body tissue accretion when replacing other energy sources such as fat and fiber (Reynolds et al., 2001; Boerman et al., 2015).

Starch from corn silage or high-moisture corn increases in digestibility as ensiling time increases (Ferraretto et al., 2015). Previous research with Enogen silage investigated effects after more than 200 d of fermentation, when starch digestibility is typically quite high (Cueva et al., 2021b). Such a long fermentation may mask benefits of enhanced starch digestibility resulting from increased enzyme activity immediately postharvest. Investigations of Enogen silage early after harvest may yield different results than those after longer fermentations. Also, improved silage fermentation may improve diet adaptation when feeds and diets are abruptly changed; Enogen may improve adaptation of lactating cows to a new diet.

An emerging area of research is investigating the effects of diet fermentability on systemic inflammation of lactating dairy cows. Ruminal grain challenge experiments have observed that bouts of ruminal acidosis may lead to systemic inflammation (Li et al., 2011). Khafipour et al. (2009b) observed that increasing fermentable starch supply of a diet with wheat and barley pellets resulted in a systemic inflammatory response likely related to ruminal or hindgut acidosis. Evaluation of systemic inflammation in chronic feeding studies is less common. Intestinal starch infusions designed to cause hindgut acidosis demonstrated that fecal pH is consistently decreased, yet neither systemic inflammatory responses nor evidence of impaired gut function were observed (van Gastelen et al., 2021a; van Gastelen et al., 2021b). These data demonstrate that increasing starch flow to the small intestine may not cause inflammation but increasing amounts of fermentable grain fed to dairy cattle may cause inflammation associated with ruminal acidosis. It also may be that sudden dietary changes lead to inflammation more than a chronically high-starch diet (Gott et al., 2015).

Therefore, the objectives of our experiment were to evaluate the effects of Enogen corn silage that had been ensiled for 41 d fed with different dietary starch contents on milk yield, component yield, plasma biomarkers of inflammation and metabolism, fecal pH, and total-tract nutrient digestibility. Our hypothesis was that the enhanced amylase activity of Enogen silage would increase starch digestibility in the rumen which would increase total-tract starch digestibility, milk yield, and milk component yields. We also hypothesized that increasing dietary starch content would increase milk production and would not affect circulating plasma acute phase protein concentrations.

MATERIALS AND METHODS

Cows, Treatments, and Design

This experiment was approved by the Michigan State University Institutional Animal Care and Use Committee. The 44 cows (28 multiparous, 16 primiparous) enrolled in this experiment were housed in individual tie stalls at the Michigan State Dairy Cattle Teaching and Research Center. Cows were fed once daily for ad libitum consumption (~10% refusal rate) at 0800 h and had ad libitum access to water. Cows were milked 3 times daily in a milking parlor at 0445, 1245, and 2045 h. Cows were let outside for exercise for approximately 1 h before each milking. At enrollment, cows averaged 151 ± 42.4 DIM with a range from 78 to 225 DIM, 668 ± 63.6 kg of BW, and 45 ± 7.3 kg/d of milk with 3.8 ± 0.51% fat and 3.1 ± 0.24% protein.

The experimental treatments were arranged in a 2 × 2 factorial layout. The first factor was corn silage and the second factor was dietary starch content. The corn silage varieties were Enogen (**ENO**) or the control silage (**CON**), which was an isoline hybrid that did not have the inserted α-amylase gene. Both corn silage varieties were included at 40% of the diet DM. Dietary starch contents were formulated at either 25% of diet DM (**LO**) or 30% of diet DM (**HI**) by replacing soybean hulls with dry ground corn. These factors created 4 treatment combinations: CON-LO, CON-HI,

ENO-LO, and ENO-HI. Additionally, diets were composed of alfalfa silage, cottonseed and grain mixes. The diets were balanced to meet nutrient requirements of dairy cows producing 50 kg/d of milk with 3.9% fat and 3.1% protein using Ag Modeling and Training Systems software (AMTS LLC.).

Silage plots (~6 ha each) were side by side and managed identically. The Enogen silage was E110F4 (Syngenta), and the control variety was G11V76 (GoldenHarvest). Both varieties were planted at 70,074 seeds/hectare on May 6, 2021, and harvested using a self-propelled Claas forage harvester equipped with a kernel processor on September 10, 2021. Silages were ensiled in bags with a *Lactobacillus buchneri* 40788 inoculant which also included *Pediococcus pentosaceus* (Crop-N-Rich Stage 2 100, VitaPlus). Bags were sealed for 41 d before feeding began.

The experiment was a randomized complete block design with cows blocked by DIM and milk yield. Treatments were randomly assigned to cows within each block (n = 11 cows per treatment). The experiment had a 2-wk covariate period where cows were fed a diet with 26% dietary starch content, 30.8% NDF, 17% CP, and a brown mid-rib corn silage hybrid. The 2-wk covariate period was followed by a 6-wk experimental period. Using variance from a previous experiment (Ylioja et al., 2018), at $\alpha = 0.05$ and a power of test $(1 - \beta)$ of 0.90, this design was expected to be sufficient to detect a change in milk yield of 3 kg/d. Covariate period summary statistics for cows enrolled on each treatment are listed in Table 1.

Data Collection and Sample Analysis

Feed intake was measured daily throughout the experiment. Milk samples were collected at all milkings during the last 4 d of the covariate period for use as parameters in statistical analysis. Milk samples were taken at all 3 milkings 1 d per week throughout the experimental period. Individual milk samples were analyzed by Central Star Cooperative for milk fat, milk true protein, and MUN concentrations using Fourier transform infrared spectroscopy and SCC by flow cytometry (Bentley FTM/FCS, Bentley Instruments Inc.). Daily milk yield was collected by an automatic milk yield recording system in the milking parlor (BouMatic).

Body weight was measured 3 times per week after the 1245 h milking. Body condition score was measured on a 1–5 scale (Wildman et al., 1982) each week, independently, by 2 trained investigators.

Blood samples were collected at 0700 h, ~1 h before feed delivery, on d 0 and weekly thereafter throughout the experimental period. Blood was collected by coccygeal venipuncture with tubes containing K_2EDTA. Blood samples were immediately placed on ice and transported to the laboratory where they were centrifuged for 15 min at 1,500 × *g* and 4°C to collect plasma. Two aliquots of plasma were kept at −20°C for later analysis. Plasma was analyzed for glucose by enzymatic methods (kit #997–03001; Fujifilm), insulin using a bovine specific ELISA (no. 10–1201–01; Mercodia AB), haptoglobin using a bovine specific ELISA (kit # HAPT-11; Life Diagnostics), and serum amyloid

Table 1. Days in milk, BW, BCS, milk yield, and milk component production of cows during the 2-wk covariate period, grouped by subsequent treatment assignment

	Treatment[1,2]			
	CON		ENO	
Item	LO	HI	LO	HI
DIM	153 ± 45	149 ± 41	160 ± 43	155 ± 46
BW, kg	660 ± 71	651 ± 71	669 ± 64	695 ± 53
BCS	3.11 ± 0.38	3.16 ± 0.30	3.23 ± 0.24	3.52 ± 0.49
Milk yield, kg/d	46.1 ± 9.4	43.6 ± 5.4	46.3 ± 7.2	44.3 ± 6.4
Milk fat, %	3.83 ± 0.43	3.66 ± 0.59	3.77 ± 0.58	3.85 ± 0.46
Milk fat yield, kg/d	1.74 ± 0.28	1.58 ± 0.25	1.72 ± 0.21	1.70 ± 0.32
Milk protein, %	3.16 ± 0.26	3.03 ± 0.24	3.07 ± 0.22	3.18 ± 0.18
Milk protein yield, kg/d	1.44 ± 0.21	1.32 ± 0.15	1.42 ± 0.19	1.40 ± 0.18
Milk lactose, %	4.89 ± 0.08	4.89 ± 0.10	4.91 ± 0.07	4.91 ± 0.10
Milk lactose yield, kg/d	2.25 ± 0.46	2.13 ± 0.26	2.27 ± 0.35	2.18 ± 0.33
SCC, ×1,000 cells/mL	34 ± 41	32 ± 33	17 ± 13	48 ± 100
MUN, mg/dL	14.4 ± 1.9	14.6 ± 1.6	14.5 ± 1.5	14.4 ± 2.0

[1]Mean ± SD, n = 11.

[2]CON = control corn silage; ENO = α-amylase enzyme enhanced corn silage (Enogen, Syngenta Seeds LLC); LO = low starch (25% of diet DM); HI = high starch (30% of diet DM). Treatments assigned after the covariate period.

A (**SAA**) using a multispecies ELISA (kit # TP 802; Tridelta Development Ltd.). The interassay coefficients of variation were 2.0, 3.1, 5.7, and 12.0 for glucose, insulin, haptoglobin, and SAA, respectively. Intraassay coefficients of variation were 6.2, 6.1, 6.0, and 7.3 for glucose, insulin, haptoglobin, and SAA, respectively.

The experiment had 2 apparent total-tract digestibility measurement periods to investigate how digestibility of the diets may have changed as ensiling time of the silages increased. Feed ingredients and diet refusal samples were collected from d 4 to d 8 and from d 39 to d 42 of the experimental period. Feed and refusal samples were composited within period. Also, fecal samples (~300 g) were caught during defecation or collected from the rectum every 9 h over d 5–8 and d 38–42 and composited by cow within measurement period. Feed ingredient samples were sent to Cumberland Valley Analytical Services and analyzed for DM (AOAC International, 2000), CP (Leco FP-528 N Combustion Analyzer, Leco Corp.), NDF (Van Soest et al., 1991), aNDFom (Mertens, 2002), in vitro NDF digestibility (Goering and Van Soest, 1970), ADF (method 973.18; AOAC International, 2000), starch (Hall, 2009), 7-h in vitro starch digestibility (Goering and Van Soest, 1970; Richards et al., 1995), sugar (DuBois et al., 1956), crude fat (2003.05; AOAC International, 2006), fatty acids (Sukhija and Palmquist, 1988), and minerals (2003.05; AOAC International, 2006). Both fecal and feed refusal samples were analyzed for DM, aNDFom, NDF, starch, and CP. We used undigested NDF after 240 h of in vitro fermentation as the internal marker to estimate fecal output. Apparent total-tract nutrient digestibility was obtained by dividing nutrient digested (nutrient intake-nutrient in feces) by nutrient intake. This was done to calculate digestibility during both wk 1 and 6.

Additional samples of both control and Enogen silages were collected 3 times per week during the experimental period and frozen at −20°C. Each sample was analyzed for fermentation metabolites and silage pH at Cumberland Valley Analytical Services. Briefly, a thawed corn silage sample (~25 g) was diluted with 200 mL of deionized water overnight. The mixture was blended and strained through 20 μm filter paper. The extract was then analyzed for pH (Mettler DL12 Titrator, Mettler-Toledo Inc.), ammonia (Labconco Rapidstill II, Labconco), lactic acid (YSI 2700 Select Biochemistry Analyzer), VFA, and other silage fermentation metabolites (Perkin Elmer AutoSystem gas chromatograph).

Additional fecal samples were collected one time each week approximately 12 h after feeding for determination of fecal pH using methods described by Branstad et al. (2017). Twelve h after feeding was selected because this is when fecal pH reached a minimum in a grain challenge model (Sulzberger et al., 2016). Briefly, fecal pH was measured immediately after sample collection measured by mixing feces (~50 g) with water at a 1:1 ratio by weight. The feces and water were homogenized with a blender (#58148A; Hamilton Beach) and strained through cheesecloth. The pH of the liquid was measured using a hand-held pH probe (PH100; Extech Instruments) calibrated according to manufacturer instructions (pH 4.0 and 7.0).

Statistical Analysis

The data were analyzed with a linear mixed model approach using the GLIMMIX procedures of SAS (9.4; SAS Institute Inc.). Models included the fixed effects of corn silage, starch, and week along with their 2- and 3-way interactions. The model also included fixed effects of parity (primiparous vs. multiparous) and pretreatment covariate along with their interactions with starch and corn silage. Block and cow were included as random effects. Corn silage, starch, week, and their interactions were always included in the model. Parity, covariate, and their interactions with starch, corn silage, and week remained in the model if $P \leq 0.10$. Model hierarchy was always maintained. Weekly averages for milk yield and DMI were calculated and used for analysis. Weekly milk yield, component yields, plasma metabolites, DMI, digestibility (wk 1 and 6), and fecal pH were evaluated as repeated measures within cow. The covariance structures were selected based on the structure that yielded the lowest Akaike information criterion for each respective variable. We evaluated compound symmetry, autoregressive, autoregressive heterogeneous, and Toeplitz covariance structures for each variable. Change in BW per day (**ΔBW**) and change in BCS per week were determined by conducting a simple linear regression for each cow. The slope coefficient from these regressions were analyzed. Statistical differences were declared at $P \leq 0.05$ and tendencies at $P \leq 0.10$. All models were visually evaluated to ensure normality of residuals. Furthermore, if an observation had a Studentized residual ≥ 4 or ≤ -4, the observation was removed and the model was re-fit.

RESULTS

Ingredient and Diet Nutrient Composition

The control and Enogen silages had similar nutrient composition (Table 2). 30-h in vitro NDF digestibility was 45% for control and 52% for Enogen silage and 7-h in vitro starch digestibility was 67 and 74% for control and Enogen, respectively, during wk 1. During wk 6, in vitro digestibilities were similar for control and Enogen

Table 2. Nutrient composition of corn silages, alfalfa haylage, whole cottonseed, high-starch grain mix, and low-starch grain mix used in an experiment feeding α-amylase-enhanced corn silage with 25 or 30% dietary starch[1]

Item,[2] % DM unless noted otherwise	Control corn silage[3]		Enogen corn silage[4]		Alfalfa haylage		Whole cottonseed		LO concentrate[5]		HI concentrate[6]	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
DM, % as-is	35.8	1.01	35.2	2.00	46.8	0.01	90.8	0.05	89.7	0.10	88.1	2.17
CP	7.4	0.71	7.8	0.07	18.3	0.57	23.3	3.25	25.9	0.28	24.6	1.06
Soluble protein	49.1	4.24	47.2	8.98	62.5	6.43	31.3	20.01	17.9	14.57	20.2	12.1
ADF	21.8	0.64	21.6	0.71	31.8	0.00	33.5	0.14	14.2	0.71	6.7	0.07
NDF	34.1	3.68	36.0	3.25	36.9	2.62	49.0	5.37	21.9	0.99	13.3	0.64
aNDFom	33.6	4.03	35.3	2.69	36.5	2.76	47.7	4.81	21.6	0.92	12.9	0.49
dNDF30, % NDF	51.5	8.82	53.7	2.4	49.5	1.5	29.7	9.36	84.6	0.99	75.5	0.51
dNDF240, % NDF	67.6	4.36	70.1	0.89	57.1	0.55	59.1	3.67	90.1	1.56	86.6	1.01
ADICP	0.5	0.07	0.6	0.02	1.4	0.02	1.8	0.85	0.6	0.15	0.4	0.05
NDICP	1.4	0.76	0.7	0.01	1.8	0.44	2.4	0.50	1.1	0.77	2.9	2.54
Lignin	2.6	0.18	3.0	0.29	5.9	0.43	11.0	0.63	1.4	0.88	0.8	0.13
Ethanol-soluble carbohydrate	1.0	0.00	2.3	1.41	4.5	0.78	4.0	1.41	5.2	1.77	5.6	0.14
Starch	37.0	0.07	37.8	1.77	1.9	0.14	0.6	0.35	24.4	2.05	34.4	2.76
7-h starch digestibility, % starch	70.4	3.54	73.0	1.48	—	—	—	—	—	—	—	—
Crude fat	2.6	0.16	2.8	0.22	2.7	0.13	16.9	0.71	2.1	0.18	3.8	2.04
Total fatty acids	2.0	0.31	2.3	0.18	1.7	0.16	16.2	0	2.4	0.13	3.9	2.31
Ash	3.1	0.58	3.2	0.28	9.8	0.79	4.7	1.32	11.0	0.60	11.4	1.60
Ca	0.2	0.06	0.3	0.06	1.8	0.24	0.2	0.02	1.9	0.23	2.0	0.45
P	0.2	0.00	0.2	0.01	0.3	0.05	0.7	0.03	0.6	0.01	0.6	0.01
Mg	0.1	0.02	0.2	0.02	0.3	0.03	0.4	0.02	0.7	0.27	0.6	0.07
K	0.8	0.04	0.7	0.01	2.0	0.21	1.3	0.03	1.4	0.13	1.4	0.04

[1]n = 2 composite feed samples.

[2]aNDFom = α-amylase treated NDF − NDF ash; dNDF30 = aNDFom digested after 30 h in vitro; dNDF240 = aNDFom digested after 240 h in vitro; ADICP = acid detergent insoluble CP; NDICP = neutral detergent insoluble CP.

[3]Isoline control corn silage.

[4]α-amylase-enhanced corn silage (Enogen, Syngenta Seeds LLC).

[5]Low-starch concentrate mix.

[6]High-starch concentrate mix.

silages. The LO and HI concentrate mixes differed in the amount of NDF and starch by design (Table 2). Diets were similar in nutrient composition except for starch and NDF concentrations (Table 3). Targeted starch concentrations of 25 and 30% for LO and HI treatments, respectively, matched well with measured values.

Silage Fermentation Analysis

Weekly silage samples collected during the 6-wk experimental period contained 6.5 ± 1.49% total VFA (DM basis; mean ± SD) for control silage and 7.0 ± 2.10% VFA for Enogen silage (Figure 1). Acetic acid was 1.2 ± 0.75% of DM and 1.6 ± 0.78% of DM for control and Enogen silages, respectively. Lactic acid was 5.3 ± 1.49% for control and 5.3 ± 1.72% for Enogen silage. Silage pH was 3.91 ± 0.089 for control and 3.93 ± 0.094 for Enogen. The 7-h in vitro starch digestibility was 66.2 ± 6.32 and 65.5 ± 4.33% of starch for control and Enogen silage, respectively.

Metabolism and Inflammation

Plasma glucose, insulin, haptoglobin, and serum amyloid A concentrations were unaffected by corn silage variety or starch concentrations (Table 4). Each biomarker was affected by week but no interactions of week with corn silage or starch were observed.

Nutrient Intake, Digestibility, and Fecal pH

A starch × week interaction was observed for DMI ($P = 0.02$; Table 5). By wk 6, cows consuming HI ate 1.8 ± 0.93 kg/d less than cows consuming LO ($P = 0.05$; Figure 2). The starch × week interaction also affected NDF, starch, and CP intake ($P < 0.01$). Neutral detergent fiber intake was greater for LO in both periods ($P < 0.01$). During wk 1, HI cows consumed 1.4 ± 0.27 kg/d more starch than LO cows ($P < 0.01$) but by wk 6, HI cows were only eating 0.6 ± 0.27 kg/d more starch than LO cows. Intake of CP during wk 1 was similar across groups ($P = 0.73$) but

Table 3. Ingredient and nutrient composition diets containing either control or α-amylase-enhanced corn silage with low (25%) or high (30%) concentrations of dietary starch

	Treatment[1]			
	CON		ENO	
Item	LO	HI	LO	HI
Ingredient, % DM				
Control corn silage[2]	40.0	40.0	—	—
Enogen corn silage[3]	—	—	40.0	40.0
Alfalfa silage	12.5	12.5	12.5	12.5
Dry ground corn	13.9	20.8	13.9	20.8
Soybean meal	11.3	11.3	11.3	11.3
Soybean hulls	9.9	3.0	9.9	3.0
Cottonseed	3.4	3.4	3.4	3.4
Heat-treated soybean meal[4]	4.9	4.9	4.9	4.9
RP-Met[5]	0.08	0.08	0.08	0.08
Mineral and vitamin premix[6]	3.6	3.6	3.6	3.6
Chemical composition,[7] % DM unless noted otherwise				
DM	52.2 (0.57)	51.9 (0.03)	51.9 (1.15)	51.5 (0.62)
CP	17.43 (0.47)	16.8 (0.19)	17.7 (0.05)	17.0 (0.61)
ADF	20.0 (0.08)	16.2 (0.78)	19.6 (0.19)	16.0 (0.34)
NDF	29.5 (2.47)	25.8 (2.50)	30.2 (2.37)	26.5 (2.45)
aNDFom	29.0 (2.58)	25.3 (2.58)	29.7 (2.10)	25.9 (2.15)
Forage NDF	17.8 (2.14)	18.0 (2.34)	18.4 (1.79)	18.6 (1.99)
dNDF30, % NDF	65.3 (3.45)	61.1 (4.05)	66.3 (0.71)	62.0 (1.35)
dNDF240, % NDF	73.1 (1.30)	69.0 (1.63)	74.2 (0.28)	70.3 (0.17)
Starch	26.0 (0.84)	30.5 (3.29)	26.3 (1.39)	30.7 (1.84)
Total fatty acids	2.6 (0.08)	3.3 (1.14)	2.7 (0.22)	3.4 (1.08)
Ash	7.4 (0.58)	7.6 (0.99)	7.5 (0.42)	7.6 (0.82)
ROM	17.0 (2.77)	16.1 (3.20)	15.6 (1.47)	14.7 (1.90)

[1]CON = control corn silage; ENO = α-amylase-enhanced corn silage; LO = low starch (25% diet DM); HI = high starch (30% of diet DM).

[2]Isoline corn silage without enhanced α-amylase.

[3]α-amylase-enhanced corn silage.

[4]Amino Plus, Ag Processing Inc.

[5]RP-Met = rumen-protected Met (Smartamine M; Adisseo).

[6]Mineral and vitamin premix composition, DM% basis: 9.6% dicalcium phosphate, 5.76% Mg oxide, 9.60% NaCl, 38.41% $CaCO_3$, 1.25% Se premix (600 mg/kg), 34.86% sodium sesquicarbonate, 0.50% vitamin/trace mineral premix.

[7]Data presented as means (SD); n = 2 composite samples; aNDFom (α-amylase treated) = NDF − NDF ash; dNDF30 = aNDFom digested after 30 h in vitro; dNDF240 = aNDFom digested after 240 h in vitro; ROM (residual organic matter) = 100 − %CP − % total fatty acids − %ash − %starch − %NDF.

by wk 6, cows receiving LO consumed 0.6 ± 0.17 kg/d more CP than cows receiving HI ($P < 0.01$).

Dry matter digestibility (**DMD**) was affected by starch × week ($P < 0.01$) and corn silage × week ($P < 0.01$) interactions. In both wk 1 and 6, HI and LO resulted in similar DMD ($P > 0.80$). During wk 1, ENO cows had 3.4 ± 0.38 percentage units greater DMD ($P < 0.01$), but by wk 6, DMD was similar across corn silages ($P = 0.22$; Figure 3). Neutral detergent fiber digestibility (**NDFD**) was affected by a week × corn silage × starch interaction. During wk 1, ENO-LO had the greatest NDFD (48.7 ± 0.95%; $P < 0.01$), followed by ENO-HI and CON-LO (42.2 ± 0.95%, 41.5 ± 0.95%; $P = 0.94$), whereas CON-HI had the least NDFD in wk 1 (33.3 ± 0.90%; $P < 0.01$). During wk 6, CON-LO and ENO-LO had similar NDFD ($P = 0.18$; 45.4 ± 0.95%, 47.5 ± 0.95%), and CON-HI and ENO-HI also had similar NDFD ($P = 0.38$; 38.6% ± 1.0%, 41.4% ± 0.95%). Cows fed LO had greater NDFD than cows fed HI during both wk 1 and 6 ($P < 0.01$). Starch digestibility was affected by starch × week ($P = 0.04$) and corn silage × week ($P < 0.01$) interactions. Within week, HI and LO starch digestibility did not differ ($P > 0.23$). Starch digestibility for LO during wk 1 was greater than starch digestibility for HI during wk 6 ($P = 0.04$). During wk 1, ENO tended to have 0.4 ± 0.02 percentage units greater starch digestibility than CON ($P = 0.07$), but during wk 6 the starch digestibility was similar for CON and ENO ($P = 0.41$). CP digestibility was also affected by the starch × week ($P < 0.01$) and corn silage × week ($P < 0.01$) interactions. In wk 1, HI and LO had similar

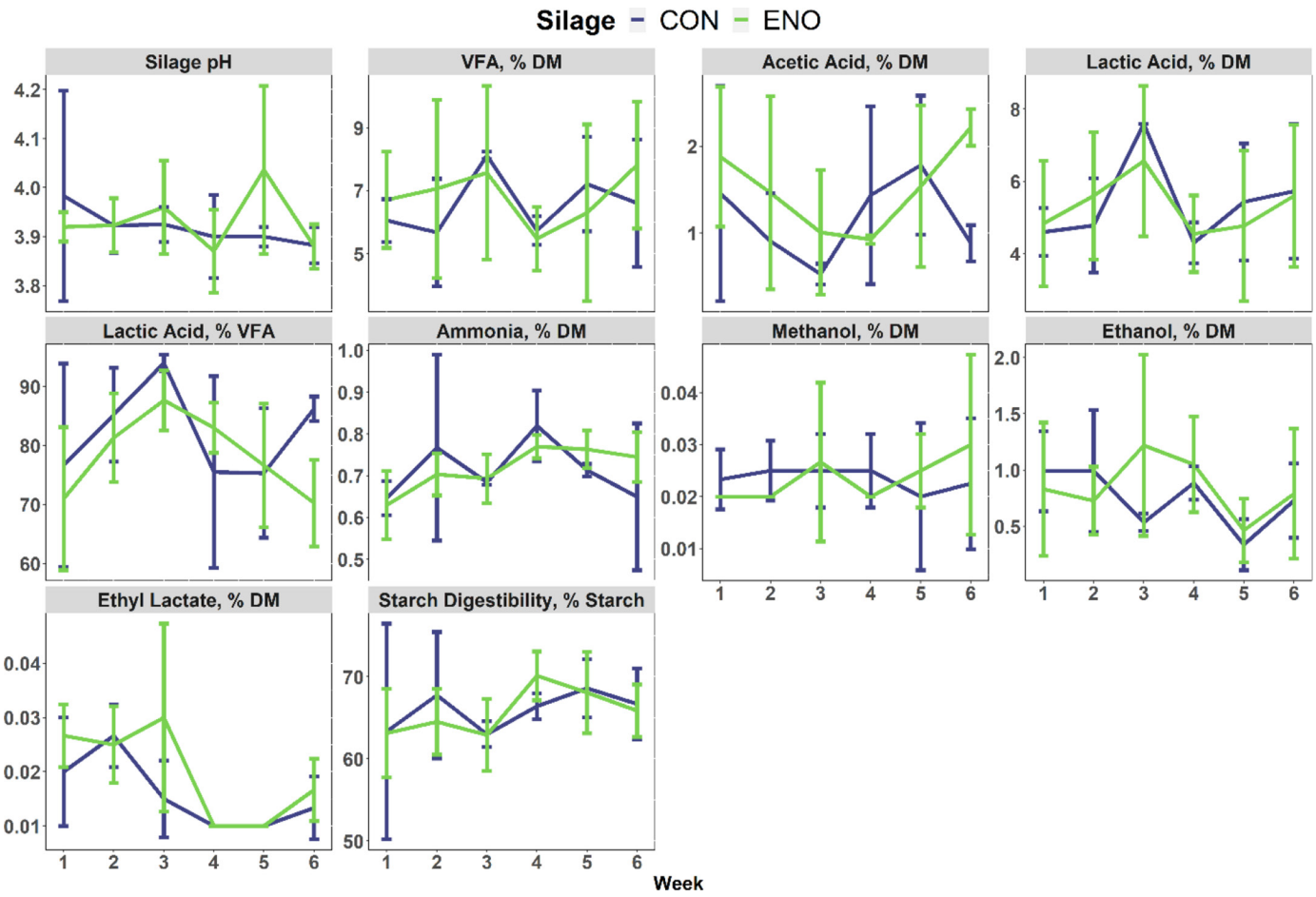


Figure 1. Silage fermentation analysis for silage (n = 18) with enhanced endogenous α-amylase (ENO; Enogen, Syngenta Seeds LLC) or a control silage without the α-amylase gene (CON). Samples were collected 3 times per week from 41 d postharvest until 83 d postharvest. Values are means ± SD.

CP digestibility ($P = 0.31$) but during wk 6, LO had 1.9 ± 0.79 percentage units greater CP digestibility than HI ($P = 0.01$). During wk 1, ENO had 4.0 ± 0.73 percentage units greater CP digestibility than CON ($P < 0.01$) but during wk 6 they were similar ($P = 0.92$).

Fecal pH (Table 5) was not affected by starch ($P = 0.75$) but it tended to be affected by corn silage

Table 4. Plasma glucose, insulin, haptoglobin, and serum amyloid A of dairy cows fed a control (CON) or α-amylase-enhanced corn silage (ENO) with low (LO; 25%) or high (HI; 30%) concentrations of dietary starch (n = 11 cows per treatment)

	Treatment					*P*-value						
	CON		ENO[1]									
Item	LO	HI	LO	HI	SEM	Corn silage	Starch	Week[2]	CS[3] × starch	CS × week	Starch × week	CS × starch × week
Glucose, mg/dL	69.8	69.7	70.9	69.5	1.39	0.66	0.45	<0.01	0.55	0.19	0.86	0.79
Insulin, μg/L	0.26	0.23	0.24	0.25	0.024	0.99	0.48	<0.01	0.34	0.29	0.77	0.76
Haptoglobin, μg /mL	6.32	4.08	3.62	6.84	3.71	0.99	0.89	0.03	0.46	0.48	0.27	0.82
Serum amyloid A, μg/mL	10.61	8.96	8.87	14.04	4.64	0.67	0.63	0.05	0.31	0.61	0.20	0.42

[1]Enogen, Syngenta Seeds LLC.
[2]Weekly data during all 6 wk of experimental period.
[3]CS = corn silage.

Table 5. Nutrient intake, nutrient digestibility, BW change, BCS, and fecal pH of dairy cows fed a control (CON) or α-amylase-enhanced corn silage (ENO) with low (LO; 25%) or high (HI; 30%) concentrations of dietary starch (n = 11 cows per treatment)[1]

	Treatment					*P*-value						
	CON		ENO[2]									
Item	LO	HI	LO	HI	SEM	Corn silage	Starch	Week[3]	CS[4] × starch	CS × week	Starch × week	CS × starch × week
Intake, kg/d												
DM	28.0	26.3	27.5	27.3	0.91	0.75	0.26	0.41	0.40	0.77	0.02	0.28
NDF	8.3	6.7	8.3	7.1	0.24	0.53	0.01	<0.01	0.41	0.53	<0.01	0.35
Starch	7.2	8.0	7.2	8.4	0.26	0.45	0.03	<0.01	0.48	0.12	<0.01	0.15
CP	4.9	4.4	4.9	4.6	0.16	0.47	0.03	0.20	0.55	0.12	<0.01	0.16
Digestibility, %												
DM	67.8	67.6	69.6	69.9	0.39	<0.01	0.82	0.17	0.56	<0.01	<0.01	0.80
NDF	43.4	35.9	48.1	41.8	0.95	<0.01	<0.01	<0.01	0.53	<0.01	<0.01	0.03
Starch	98.0	97.7	98.2	98.1	0.23	0.18	0.34	<0.01	0.67	<0.01	0.04	0.69
CP	66.8	67.1	69.7	68.4	0.79	0.01	0.39	<0.01	0.18	<0.01	<0.01	0.79
ΔBW,[5] kg/d	0.49	0.35	0.43	0.23	0.094	0.27	0.05	—	0.74	—	—	—
ΔBCS,[6] units/wk	−0.009	−0.015	−0.006	−0.015	0.0097	0.84	0.46	—	0.87	—	—	—
Fecal pH	6.76	6.82	6.87	6.83	0.032	0.06	0.75	0.05	0.15	0.36	0.71	0.49

[1]Means presented as LSM across both wk 1 and 6 digestibility.
[2]Enogen, Syngenta Seeds LLC.
[3]Weely data during wk 1 and 6 except for fecal pH where data are from all 6 wk were collected.
[4]CS = corn silage.
[5]Change in BW per day.
[6]Change in BCS per week.

($P = 0.06$), with an increase of 0.07 ± 0.033 units for ENO compared with CON.

BW and BCS

The change in BW was greater for LO than HI ($P = 0.05$; Table 5), whereas corn silage had no effect on ΔBW. Change in BCS per week was not affected by corn silage, starch, or the interaction ($P > 0.46$).

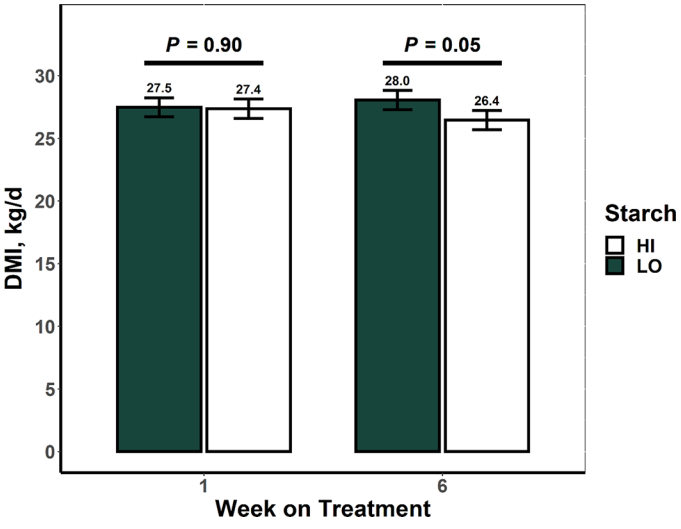


Figure 2. Dry matter intake during wk 1 and 6 of cows fed α-amylase-enhanced corn silage (Enogen, Syngenta Seeds LLC) or a control silage without the α-amylase gene with low (LO; 25% of diet DM) or high (HI; 30% of diet DM) concentrations of dietary starch (n = 11 cows per treatment). Values are LSM ± SEM.

Milk Yield and Milk Composition

Milk yield tended to be greater for HI than LO ($P = 0.10$; Table 6) but was not affected by corn silage ($P = 0.38$). Energy-corrected milk yield also tended to be greater for HI than LO ($P = 0.07$) and was not affected by corn silage ($P = 0.21$). Feed efficiency (**FE**), calculated as ECM per DMI, tended to be greater for HI than LO ($P = 0.07$) and tended to be affected by the interaction of corn silage × week ($P = 0.09$), such that cows receiving ENO had reduced FE during wk 6 compared with wk 1 ($P < 0.01$) whereas cows fed CON had similar FE in both wk 1 and 6 ($P = 0.75$). Within week, cows consuming CON or ENO had similar FE ($P > 0.65$) in both wk 1 and 6.

Milk fat concentration tended to be affected by a corn silage × starch interaction ($P = 0.06$) but after Tukey adjustments none of the treatments differed from each other ($P \geq 0.27$). Fat yield was only affected by week ($P < 0.01$). Milk protein concentration tended to be affected by a starch × corn silage × week interaction ($P = 0.10$). The CON-LO group tended to have greater milk protein content than ENO-HI and CON-HI during wk 6 ($P < 0.08$). Milk protein yield was affected by the interaction of starch × week ($P = 0.01$) such that during wk 3 ($P < 0.01$), 4 ($P = 0.04$), and 5 ($P = 0.04$), protein yield was greater for cows fed HI than cows fed LO by 142 ± 41.6, 89 ± 41.8, and 92 ± 44.6 g/d, respectively (Figure 4). Milk lactose concentration was affected by corn silage × week ($P = 0.06$) and starch × week ($P = 0.04$) interactions. During wk 2 and 3,

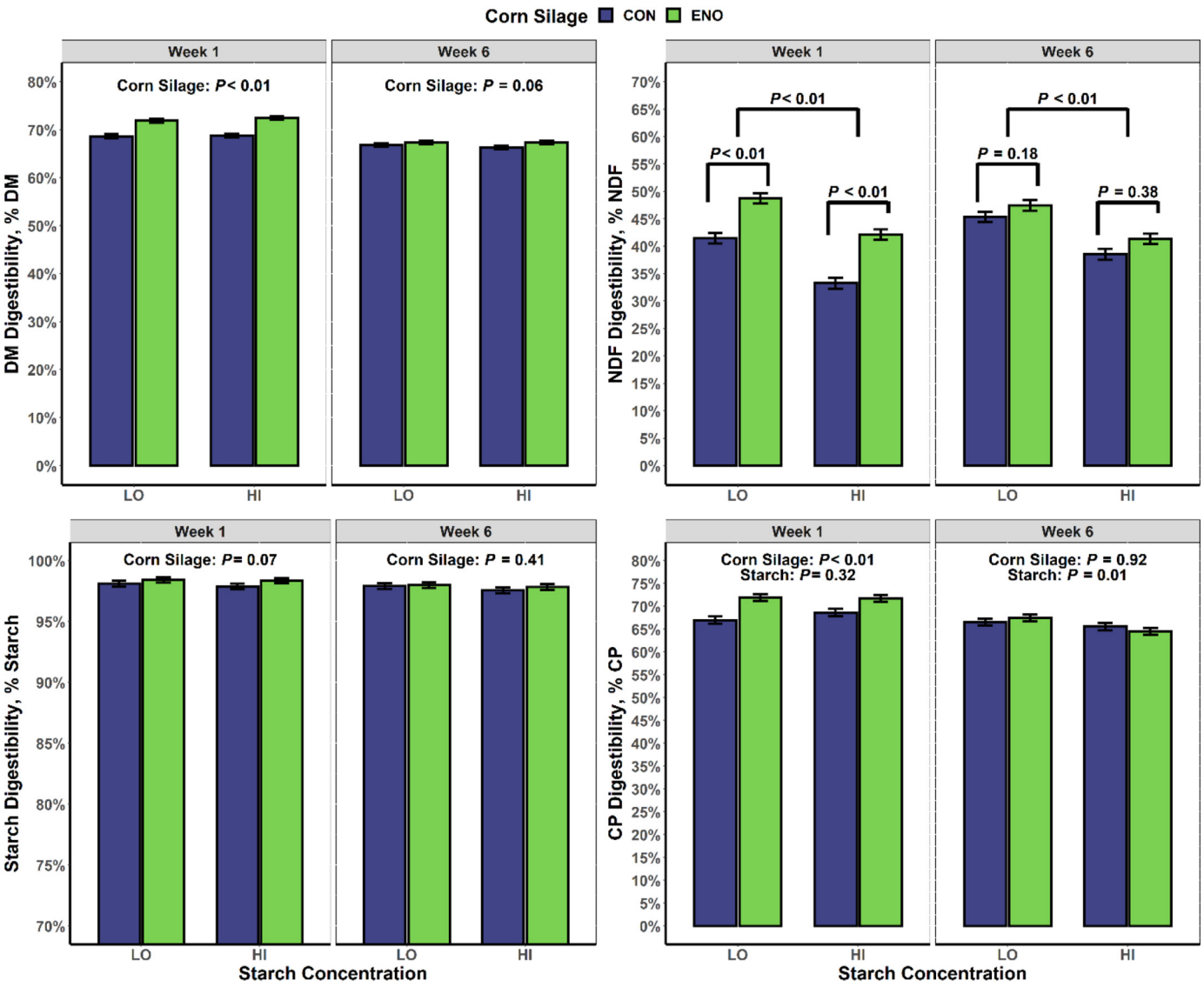


Figure 3. Apparent total-tract DM, NDF, starch, and CP digestibility during wk 1 and 6 from cows fed α-amylase-enhanced corn silage (ENO; Enogen, Syngenta Seeds LLC) or a control silage without the α-amylase gene (CON) with low (LO; 25% of diet DM) or high (HI; 30% of diet DM) concentrations of dietary starch (n = 11 cows per treatment). Values are LSM ± SEM.

CON tended to increase lactose content, whereas during wk 6, ENO tended to increase lactose content ($P = 0.09$). The starch × week interaction was such that HI had greater lactose content than LO during wk 5 and 6 ($P < 0.05$). Lactose yield was affected by starch ($P < 0.01$); HI cows yielded 104 ± 3.6 g/d more lactose than LO. Milk urea nitrogen was affected by a starch × corn silage × week interaction ($P = 0.02$). During wk 1, CON-LO tended to have greater MUN than ENO-LO ($P = 0.08$) and CON-LO had greater MUN than CON-HI and ENO-LO during wk 3 ($P < 0.03$). In wk 6, ENO-LO and CON-LO had greater MUN than CON-HI ($P < 0.01$). Furthermore, CON-LO and ENO-LO had greater MUN than ENO-HI during wk 6 ($P < 0.01$). Somatic cell score was not affected by treatments ($P > 0.15$).

DISCUSSION

Investigations of α-amylase-enhanced corn in finishing beef cattle have revealed slight increases in gain:feed ratio when fed as dry rolled corn but not when fed as high-moisture corn (Volk et al., 2021). To date, Cueva et al. (2021b) and Rebelo et al. (2020) have published the only 2 experiments investigating this trait in corn silage for dairy cattle diets. Previous results in dairy cattle are equivocal, as Enogen silage has increased milk yield but not energy-corrected milk yield, slightly

Table 6. Milk yield, milk composition, and milk component yield of dairy cows fed a control (CON) or α-amylase-enhanced corn silage (ENO) with low (LO; 25%) or high (HI; 30%) concentrations of dietary starch (n = 11 cows per treatment)

	Treatment					*P*-value						
	CON		ENO[1]									
Item	LO	HI	LO	HI	SEM	Corn silage	Starch	Week[2]	CS[3] × starch	CS × week	Starch × week	CS × starch × week
Milk yield, kg/d	38.6	40.6	38.2	39.5	0.91	0.38	0.10	<0.01	0.70	0.86	0.14	0.37
ECM, kg/d	41.3	42.7	40.5	41.7	0.77	0.21	0.07	<0.01	0.86	0.87	0.38	0.68
FE,[4] ECM/DMI	1.47	1.58	1.49	1.54	0.042	0.79	0.07	<0.01	0.44	0.09	0.28	0.59
Milk fat, %	3.94	3.81	3.84	3.79	0.060	0.84	0.87	<0.01	0.06	0.11	0.98	0.34
Milk fat, kg/d	1.51	1.53	1.46	1.48	0.039	0.13	0.57	<0.01	0.95	0.76	0.44	0.70
Milk protein, %	3.33	3.28	3.28	3.31	0.028	0.64	0.76	<0.01	0.18	0.35	<0.01	0.10
Milk protein, kg/d	1.27	1.34	1.24	1.30	0.028	0.23	0.03	<0.01	0.93	0.23	0.01	0.18
Milk lactose, %	4.88	4.89	4.88	4.88	0.011	0.69	0.20	<0.01	0.49	0.06	0.04	0.49
Milk lactose, kg/d	1.87	2.00	1.86	1.94	0.039	0.20	<0.01	<0.01	0.49	0.79	0.51	0.64
MUN, mg/dL	13.7	12.4	12.6	12.3	0.28	0.05	<0.01	<0.01	0.08	0.01	<0.01	0.02
SCS[5]	2.80	2.71	3.32	2.85	0.260	0.15	0.28	<0.01	0.47	0.21	0.35	0.43

[1]Enogen, Syngenta Seeds LLC.

[2]Weely data during all 6 wk of experimental period.

[3]CS = corn silage.

[4]FE = feed efficiency.

[5]SCS = $\text{Log}_2(\text{SCC}/10{,}000) + 3$.

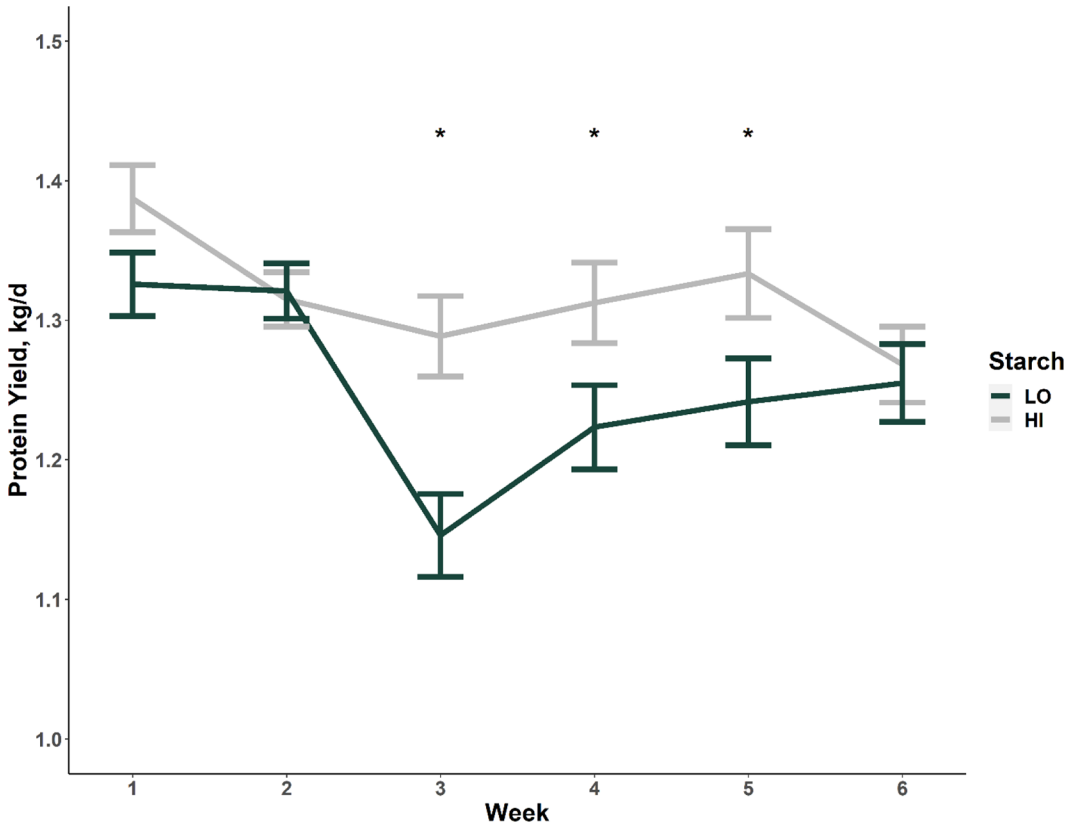


Figure 4. Milk protein yield from cows fed α-amylase-enhanced corn silage (Enogen, Syngenta Seeds LLC) or a control silage without the α-amylase gene with low (LO; 25% of diet DM) or high (HI; 30% of diet DM) concentrations of dietary starch (n = 11 cows per treatment). Milk protein yield was greater for HI than LO during wk 3, 4, and 5 ($P < 0.05$). Values are LSM ± SEM. Asterisks indicate significant differences within that week ($P < 0.05$).

reduced methane intensity and slightly increased FE (Rebelo et al., 2020; Cueva et al., 2021b). However, neither experiment evaluated Enogen silage when fed with different concentrations of dietary starch. This may be important, as increased ruminal starch digestibility puts cows at increased risk of developing subacute ruminal acidosis, especially in diets with greater levels of starch (Humer et al., 2018).

Both silage sources had silage pH <4.0 at the start of feeding, which is considered an indicator of adequate fermentation (Kung et al., 2018). In vitro starch digestibility (7-h) was also similar for both hybrids. Previous research has reported similar pH between α-amylase-enhanced and control silages (Hellings et al., 2019; Cueva et al., 2021b).

Metabolism and Inflammation

We did not observe any changes in plasma glucose or insulin concentrations. Previous research found that increasing dietary starch content by removing grass hay and increasing corn silage increased plasma glucose in early lactation (~30 DIM) but not later lactation (~190 DIM) cows (Piccioli-Cappelli et al., 2014). Insulin increased in response to greater starch supply in both early and late-lactation cows in that experiment. Similarly, Boerman et al. (2015) increased starch from 16 to 32% of the diet DM by removing alfalfa silage and increasing high-moisture corn, and observed similar circulating glucose concentrations but increased insulin for cows fed greater starch. More modest increases in starch (27 vs. 21%) provided by dry ground corn did not affect circulating glucose concentration (Sánchez-Duarte et al., 2019). Providing additional starch (high-moisture corn, corn silage) in place of forage fiber can increase glucose supply, likely from increased propionate production in the rumen, although it does not consistently increase regulated plasma glucose concentrations. When starch replaces nonforage fiber, such as soybean hulls as we did in our experiment, the effect seems less dramatic, which aligns with the phenomenon that feeding nonforage fiber yields more propionate than forage NDF (Clark and Armentano, 1997; Mullins et al., 2010; Sullivan et al., 2012).

Insulin promotes tissue accretion, which can promote BW gain in cows fed diets with greater starch content (Reynolds et al., 2001; Piccioli-Cappelli et al., 2014; Boerman et al., 2015). In this study we did not observe increased insulin concentrations or BW in cows fed the HI diet; in fact, we observed the opposite, as ΔBW was greater for cows fed LO versus HI. Weiss (2019) also observed greater ΔBW when reduced starch diets were fed; he hypothesized that metabolizable AA supply limited milk yield in reduced starch diets leading to energy partitioning to tissue. Feeding starch in place of fiber increases net portal appearance of total and essential amino acids and plasma concentrations of His, Pro, and Asn (Cantalapiedra-Hijar et al., 2014). Milk synthesis requires both energy and protein, and subtle deficiencies in AA supply that decrease expression of proteins involved in milk lactose or fatty acid synthesis may result in decreased milk energy secretion, reduced utilization of substrates from the bloodstream, and greater substrate supply to other tissues. The effect of starch concentration on BW in the current study may be a reflection of dietary effects on metabolizable AA supply and downstream repartitioning of nutrients, albeit minor. Alternatively, given the lack of corresponding treatment effects on change in BCS, rumen or gut fill may have been partially responsible for differences in ΔBW as LO had greater DMI and NDF intake than HI. When forage fiber content increases, weight of rumen contents also increased (Shaver et al., 1988), but in this case we increased dietary NDF by replacing corn grain with nonforage fiber which may not accumulate in the rumen or hindgut as forage fiber does.

Inflammation has gained greater attention in dairy cows (Bradford et al., 2015) because greater concentrations of biomarkers of inflammation are associated with poorer health and performance (Bertoni et al., 2008). This phenomenon has been tied to diet formulation, as experimentally induced ruminal acidosis results in increased markers of inflammation (Khafipour et al., 2009b; Li et al., 2011). Using a meta-analytic approach, Zebeli et al. (2012) determined that cows consuming barley- and wheat-based diets with >44% of diet DM as concentrates or <39% NDF may be at greater risk of inflammation. If these results were applicable to corn-based diets, it would present challenges for feeding high-producing cows because concentrate feeding is among the most efficient ways to provide digestible energy to dairy cows. Starch is particularly useful because it increases milk and milk protein yields as well as FE (Boerman et al., 2015; Morris et al., 2020). The cutoffs derived by Zebeli et al. (2012) suggest that each of our 4 diets were at risk of causing systemic inflammation (≤30% NDF and 48% concentrates). However, we found no evidence of inflammatory responses to feeding starch (from dry ground corn and corn silage) up to 30% of diet DM. Haptoglobin and SAA are widely used acute phase protein biomarkers of inflammation, and the concentrations reported herein are well below thresholds used to identify subclinical or clinical inflammatory conditions (Ceciliani et al., 2012; Quanz et al., 2022).

Even though grain challenge acidosis induction models lead to inflammation, our data demonstrate that chronically high-starch diets do not necessarily result

in inflammation. Further supporting this notion, Gott et al. (2015) fed cows a 28% starch diet or an acidosis induction diet in the same experiment. They observed that SAA was similar between the control (24% starch) and 28% starch diets. When cows were fed a SARA induction diet (32% starch; wheat replaced wet corn gluten feed) for 2 d, plasma SAA concentrations doubled. We speculate that chronically high-starch diets may not increase biomarkers of inflammation, but that sudden dietary changes that rapidly increase fermentable substrate in the rumen may result in inflammation. Further supporting this hypothesis, Khafipour et al. (2009a) induced ruminal acidosis by gradually increasing the supply of alfalfa pellets over 5 weeks, which did not result in an inflammatory response; in contrast, a sudden grain challenge model elicited an inflammatory response (Khafipour et al., 2009b). Additional research in a wider variety of diets with a broader set of measurements related to inflammation (APP, cytokines, gut histology and integrity measures) is warranted to determine what concentrations or sources of starch may result in systemic inflammation.

Nutrient Intake and Digestibility

Nutrient digestibility is affected by dietary nutrient concentrations and DMI (de Souza et al., 2018). It is well documented that feeding increased dietary starch content reduces the total-tract NDFD of diets fed to dairy cattle (Ferraretto et al., 2013; Boerman et al., 2015; Dias et al., 2018). In the most recent revision of the nutrient requirements of dairy cattle (NASEM, 2021), NDFD is modeled as a function of DMI and dietary starch content. For each unit increase in dietary starch content above 26% of DM, the model estimates that NDFD will drop by 0.6 percentage units. Directionally, our results agree with this model, but we observed double the reduction in NDFD predicted by NASEM (2021). It may be that the corn sources fed in our experiment were more rumen degradable than the average of the NASEM (2021) database, which may depress NDFD more than starch degraded postruminally. The committee acknowledged that ruminal starch degradability likely affects this relationship between NDFD and dietary starch content. Ferraretto et al. (2013) observed this phenomenon in a meta-analysis as ensiled corn grain reduced NDFD more than dry or steam-flaked corn sources.

Increasing starch concentration has resulted in no change (Boerman et al., 2015), reduced (Morris et al., 2020), or even increased starch digestibility (Sánchez-Duarte et al., 2019). de Souza et al. (2018) reported from a meta-regression that each 1-unit increase in dietary starch content results in a 0.19-unit decrease in starch digestibility. They also observed regional differences; observations from Georgia exhibited a more negative relationship between starch concentration and digestibility than those from Ohio and Michigan. Dietary starch content effects on starch digestibility appear very reliant on the digestibility of the starch source being used (Ferraretto et al., 2013), such that increasing supply of highly fermentable sources (e.g., ensiled or very finely ground corn) may be positive, whereas increasing the amount of starch from less fermentable sources (e.g., whole, cracked, or coarsely ground corn) may reduce starch digestibility.

We observed a slight increase ($0.4 \pm 0.02\%$) in starch digestibility for ENO versus CON during wk 1, but during wk 6 both in vivo and in vitro starch digestibility were similar for CON and ENO. This may be because longer ensiling time increases starch digestibility (Ferraretto et al., 2015) which may be enough to wash out any additional effect of the enhanced α-amylase activity. Hellings et al. (2019) observed that starch digestibility of silage with the α-amylase trait was 20 percentage units greater than its counterpart before ensiling, but after ensiling it was only 2 percentage units greater. Cueva et al. (2021b) observed similar in vitro starch digestibility between α-amylase-enhanced corn silage and the control silage; even though α-amylase activity remained 14 times greater in α-amylase-enhanced silage after >200 d of ensiling (Cueva et al., 2021b), the ensiling process increases the digestibility of starch in corn silage and likely reduces the starch digestibility advantage of having additional α-amylase activity.

Exogenous α-amylase supplementation (as opposed to genetic integration of α-amylase in corn grain) demonstrates slightly increased total-tract starch digestibility (Pech-Cervantes et al., 2022), but average starch digestibility was lower across studies (~95%) than we observed here. Exogenous amylase may enhance ruminal starch digestion, but hindgut starch digestion may largely compensate for differences in ruminal starch digestion (Nozière et al., 2014). Additionally, the rumen microbiota has native amylase activity, and the addition of Enogen silage or exogenous amylase does little to affect total ruminal amylase activity (Hristov et al., 2008; Cueva et al., 2021b). Enogen silage may improve starch digestibility early after harvest (<45 d), but it seems this effect is diminished after at least 80 d of ensiling.

A recent meta-analysis observed that exogenous amylase does not consistently affect NDFD (Pech-Cervantes et al., 2022), even though some individual experiments have observed substantial increases in NDFD when amylase was fed (Klingerman et al., 2009; Gencoglu et al., 2010; Weiss, 2019). Previous experiments with Enogen silage have not reported

effects on NDFD (Rebelo et al., 2020; Cueva et al., 2021b), but in both cases silages were ensiled longer before feeding, which may have removed the advantage of Enogen corn silage that we observed in our experiment. The reasons for the enhanced total-tract NDFD observed here, and previously with exogenous α-amylase, are not entirely clear. We, and Tricarico et al. (2008), speculate that α-amylase may increase ruminal concentrations of starch hydrolysis products maltose, dextrin, and glucose, which increase ruminal fibrolytic capacity through increased bacterial populations or enhanced bacterial function. Cueva et al. (2021a) observed a reduction in *Rikinella* and *Bifidobacterium* when Enogen silage was fed demonstrating the rumen microbiome may play a role in the observed responses. Exogenous amylase may increase growth of *Butyrivibrio fibrisolvens*, *Selenomonas ruminantium*, or *Megasphaera elsdenii* (Tricarico et al., 2008). Metagenomic or metatranscriptomic methods may be useful in determining if exogenous amylase or Enogen silages alter the functional capacity of the rumen microbiome, especially capacity for NDF digestion.

Crude protein digestibility was also greater for ENO early after harvest with diminished effects over time; we detected no effect of ENO on CP digestibility during wk 6. It may be that enhanced ruminal starch digestibility during wk 1 reduced the amount of starch entering the small intestine which limited hindgut microbial protein production and decreased microbial protein in feces. This is in line with the tendency for increased fecal pH when ENO was fed during our experiment. Exogenous α-amylase does not appear to affect CP digestibility (Pech-Cervantes et al., 2022).

Milk Yield and Milk Composition

Even though we observed increased DM digestibility with ENO and similar DMI for each silage, there was not an increase in milk yield or ΔBW when ENO was fed. Previous research demonstrated that Enogen silage increased milk yield but not ECM (Rebelo et al., 2020). Cueva et al. (2021b) observed a 2 kg/d increase in milk yield ($P < 0.01$) and a 1.4 kg/d numerical increase in ECM ($P = 0.12$). Feed efficiency was similar for ENO and CON in our study, which is contrary to previous results where FE was increased by 3% (Cueva et al., 2021b); in that experiment, however, the Enogen silage had greater starch concentration than control, and greater starch concentration resulted in greater FE in our study. Greater dietary starch content does not always result in increased FE (Boerman et al., 2015; Dias et al., 2018), and is likely dependent on what starch is substituted for and whether it changes DMI. In our experiment, starch replaced nonforage fiber which had limited effects on gut fill, but when starch replaces forage fiber it would reduce the filling effect of the diet and lead to increased DMI, as observed by Boerman et al. (2015) and Dann et al. (2015). It also may depend on ruminal starch degradability, as more rumen degradable starch sources depress intake as a result of the hypophagic effects of ruminal propionate (Allen and Bradford, 2012; Albornoz and Allen, 2018).

Even though increasing dietary starch concentration to 30% of diet DM does not consistently affect gross measures of FE, it does often increase N use efficiency (Huhtanen and Hristov, 2009). In our study, increasing dietary starch content reduced MUN. It also increased milk protein yield despite decreased CP intake which demonstrates greater N use efficiency. Morris et al. (2020) fed starch in place of fat to late-lactation Jersey cows and observed that milk N as a proportion of N intake increased from 29 to 32%. Dias et al. (2018) and Oba and Allen (2003a) also observed increased milk protein concentration and yield with additional starch. Starch likely promotes microbial protein production in the rumen by providing more fermentable organic matter than fiber (Oba and Allen, 2003b); in turn, microbial protein can increase metabolizable AA supply for milk protein synthesis. Greater starch concentrations may also increase milk protein through insulin signaling, which has been shown to influence milk protein synthesis (Griinari et al., 1997; Menzies et al., 2009). We did not observe increased pre-feeding insulin concentrations in response to increased starch, it is possible that insulin may have been greater after meal bouts (Oba and Allen, 2003a) in cows fed HI diets.

Feeding diets with greater dietary starch content or increasing rumen fermentability of starch are risk factors for milk fat depression (Ramirez Ramirez et al., 2015). Milk fat depression is caused by the incomplete biohydrogenation of polyunsaturated fatty acids which leave the rumen and downregulate genes related to milk fat synthesis in the mammary gland (Osorio et al., 2016). We observed a tendency for a silage × starch interaction for milk fat content; fat concentration was greater for cows fed CON-LO. However, milk fat yield was similar for all treatments. Diets likely did not dramatically alter rumen metabolism and biohydrogenation to induce milk fat depression. Our diets were formulated for adequate fNDF, which aids in maintaining greater rumen pH and milk fat content (Li et al., 2020), and low concentrations of polyunsaturated fatty acids (1.5–1.6% DM). Furthermore, the additional starch in HI was from dry ground corn, which is not as fermentable as ensiled corn (Ferraretto et al., 2013), minimizing risk of milk fat depression.

CONCLUSIONS

Feeding Enogen silage early after harvest increased total-tract nutrient digestibility, but the increment in nutrient digestibility diminished with time, which aligned with the in vitro digestibility results. However, ENO did not alter milk or milk component yields. Increasing dietary starch content from 25 to 30% of diet DM reduced NDFD but still resulted in increased milk and milk protein yields, likely from improved N utilization and increased energy supply. Increasing the concentration of starch did not alter plasma acute phase protein concentrations, providing no evidence of systemic inflammation. Also, increasing dietary starch content did not reduce fecal pH, suggesting minimal effects on hindgut fermentation.

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